

Partial Correlation Calculator (PCC)

Usage Guide

By Matthew Dinwiddie, Ann Marie Weidemann

I. What is the PCC?

Partial correlation is an analytic technique that facilitates measurement of the magnitude of association between two variables, while controlling for one or more extraneous variables.

The Partial Correlation Calculator (PCC) is an R Shiny application that leverages the `ppcor` R package to compute partial correlation coefficients and p-values, with added functionality for confidence intervals. The graphical user interface (GUI) allows the upload of a data set, variable selection, specification of significance level, and of the correlation type (Pearson, Spearman, or Kendall's Tau). The resulting output is publication-friendly; the inference and associated statistical methods can be downloaded as text files and directly inserted into a manuscript.

II. PCC User Interface and Operation

The PCC graphical user interface (GUI) consists of a *Calculator* and an *Information* panel.

The screenshot displays the 'PCC | Partial Correlation Calculator' interface. The top navigation bar includes 'PCC | Partial Correlation Calculator', 'Calculator', and 'Information'. The main panel is divided into two sections: 'Calculator' on the left and 'Analysis Output' on the right.

Calculator Panel:

- 1.** Select correlation type: A dropdown menu showing 'Spearman (Default)'. An information icon (i) is to the right.
- 2.** Select significance level: A dropdown menu showing '0.05 (Default)'. An information icon (i) is to the right.
- 3.** Upload dataset (.csv, .xls, .xlsx): A button labeled 'Browse...' and a text box showing 'No file selected'. An information icon (i) is to the right.
- 4.** My data has headers: A checked checkbox.
- 5.** Choose X and Y Variables: A text input field. An information icon (i) is to the right.
- 6.** Choose Z Continuous Variable(s): A text input field. An information icon (i) is to the right.
- 7.** Choose Z Categorical Variable(s): A text input field. An information icon (i) is to the right.
- 8.** Analyze: A dark button.
- 9.** Reset: A light button.
- 10.** Output available for download following analysis.

Analysis Output Panel:

- 11.** Unadjusted Correlation: 11.
- 12.** Partial Correlation: 12.
- 13.** Uploaded Data: 13.

Figure 1. **Calculator** panel of the PCC.

The Calculator panel offers the PCC's primary functionality. Here, users may upload their data and conduct a correlation analysis. Figure 1 is labeled with the various fields of the Calculator panel. Each field is accompanied by an information (i) indicator to help the user understand the fields while they are operating the PCC. More detailed field descriptions are listed below.

1. **Select correlation type:** Here, users may specify between the Spearman, Pearson, and Kendall's Tau correlation coefficients, for computation by the PCC.
 - a. **Spearman (Default):** Ideal for ordinal data or continuous data that do not meet the assumptions of Pearson's correlation.
 - b. **Pearson:** Best used when both variables are interval or ratio scale, linearly related, and normally distributed.
 - c. **Kendall's Tau:** Preferred for smaller datasets or when data contain many tied ranks; useful for ordinal data or non-parametric statistics.

2. **Select significance level:** Here, users may specify the level of statistical significance for which they wish to conduct their analysis. Options are 0.10, 0.05, and 0.01.
 - a. **0.10:** A more lenient threshold, allowing a 10% chance of committing a Type I error (rejecting a true null hypothesis).
 - b. **0.05 (Default):** The most commonly used significance level in research, balancing the risks of Type I and Type II errors. It indicates a 5% chance of rejecting the null hypothesis when it's actually true.
 - c. **0.01:** A strict threshold, allowing only a 1% chance of a Type I error. It demands strong evidence before rejecting the null hypothesis.
3. **Upload dataset (.csv, .xls, .xlsx):** Users may upload files of the appropriate extension for analysis. Files with extensions not in the list of acceptable inputs will not load.
4. **My data has headers:** Checkbox for whether your uploaded dataset has a header row, also known as column names.
5. **Choose X and Y variables:** Field that auto-populates with the columns of your uploaded file. Use it to specify the main variables (X and Y) of your analysis.
6. **Choose Z Continuous Variable(s):** Field that auto-populates with the columns of your uploaded file. Use it to specify the continuous variable(s) for which you seek to adjust.
7. **Choose Z Categorical Variable(s):** Field that auto-populates with the columns of your uploaded file. Use it to specify the categorical variable(s) for which you seek to adjust.
8. **Analyze:** Select this button once specifying analysis parameters to generate output.
9. **Reset:** Erase analysis parameters and output to reload the PCC with a clean slate.
10. **Download:** Available after selecting “Analyze”; downloads the results of your analysis.
11. **Unadjusted Correlation:** A summary of the standard (unadjusted) correlation analysis. Contains a description of the variables correlated, and their correlation coefficient’s magnitude, type, confidence interval, and p-value. *This is a standard correlation, and can be thought of as a baseline with which to compare the partial correlation.*

12. Partial Correlation: A summary of the partial correlation analysis. Contains a description of the variables correlated, and their correlation coefficient's magnitude, type, confidence interval, and p-value. *As the partial correlation, this output describes the relationship between your variables, adjusting for the impact of extraneous variables.*

13. Uploaded Data: A preview of the data uploaded, showing just the columns included in the analysis. Used to validate that uploaded data was ingested properly and that the user specified columns correctly for analysis.

See Figure 2 for an example of the Calculator pane after an analysis has been conducted and output is generated.

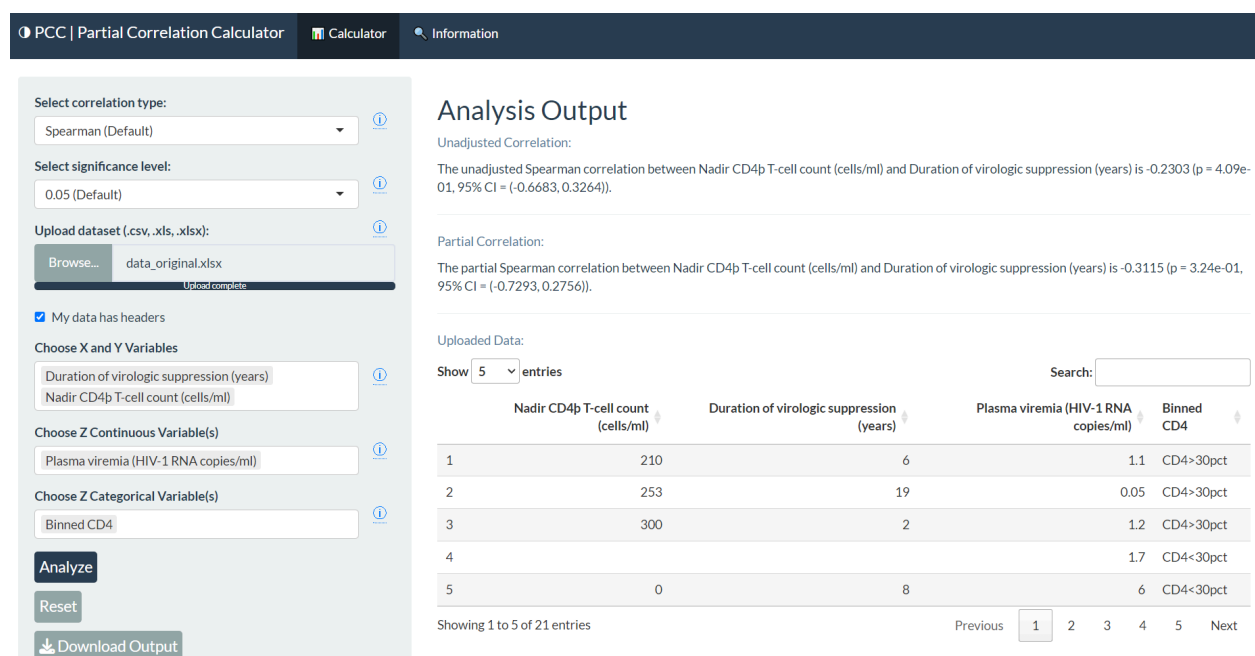


Figure 2. **Calculator** panel of the PCC, populated with analysis output.

The information panel, shown on Figure 3, contains details regarding the calculator's creators, resources that enabled the calculator, as well as a link back to this usage guide.

About this Calculator

This calculator facilitates the computation of partial correlation coefficients. Partial correlation allows researchers to quantify the strength of an association between two variables, while controlling for a set of extraneous variables.

Calculator Operation

Usage of the calculator is fairly straightforward; select from the prompts to define the parameters of your partial correlation calculation. Hover over prompts that you are unsure of to view helpful tooltips. To use this calculator, please upload your data in .CSV, .XLS, or .XLSX format. Be mindful of missingness in your data; this may hurt the quality of the analysis.

Sample Data

If you would like to test this calculator's functionality but lack data, visit [\[this link\]](#) to download sample data to use with this calculator.

Creditation

This calculator was created by Matthew Dinwiddie, MSc, and Ann Marie Weideman, MSc, in association with the University of North Carolina at Chapel Hill's Centers for AIDS Research.

References

Various R packages were utilized as well, cited below.

- Kim S. (2015). ppcor: An R Package for a Fast Calculation to Semi-partial Correlation Coefficients. Communications for statistical applications and methods, 22(6), 665–674. <https://doi.org/10.5351/CSAM.2015.22.6.665>
- Thomas J. DiCiccio. Bradley Efron. 'Bootstrap confidence intervals.' Statist. Sci. 11 (3) 189 - 228, August 1996. <https://doi.org/10.1214/ss/1032280214>
- R Core Team (2022). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. URL <https://www.R-project.org/>.

*Figure 3. **Information** containing data about the calculator's origins, usage, and references.*

III. Interpreting Results

To understand interpretation of correlation coefficients, please consult Table 1.

Table 1. Interpreting Correlation Coefficients

Correlation Range	Strength	Direction	Interpretation
-1.00 to -0.75	Strong	Negative	Very strong inverse relationship; as one variable increases, the other tends to decrease significantly.
-0.75 to -0.50	Moderate	Negative	Clear negative association, though less than very strong; inverse trend is present but with more variability.
-0.50 to -0.25	Weak	Negative	Some negative association; trend exists but not consistently strong.
-0.25 to 0.00	Very weak	Negative	Minimal to no real correlation; data may be unrelated or relationship is negligible.
0.00 to 0.25	Very weak	Positive	Very weak or negligible correlation; limited practical relationship.
0.25 to 0.50	Weak	Positive	Small but noticeable positive trend; variables move together to some extent.
0.50 to 0.75	Moderate	Positive	Moderate positive correlation; a clearer upward trend between variables.
0.75 to 1.00	Strong	Positive	Very strong direct relationship; as one variable increases, the other does too, with little deviation.

The correlation coefficients are accompanied by confidence intervals. A **confidence interval** provides a range of values within which the **true population correlation** is likely to fall, based on your sample data, with a given level of confidence (typically 90%, 95%, or 99%).

- **The wider the interval**, the **less precise** the estimate, often due to small sample sizes or high variance.
- **The narrower the interval**, the **more precise** the estimate, indicating more confidence in the correlation's stability.

Table 2 describes interpreting the results of confidence intervals at varying levels of confidence.

Table 2. Interpreting Confidence Intervals by Confidence Level

Confidence Level	Interpretation
90% CI	This interval was calculated so that, in repeated sampling, 90% of such intervals would capture the true population correlation. Often used in exploratory research where flexibility is acceptable.
95% CI	A common choice. This method produces intervals that would include the true population correlation 95% of the time across many samples. Balances precision with reliability.
99% CI	A stricter threshold, producing wider intervals that are more likely to contain the true value. Often used when the consequences of error are high.

Lastly, Table 3 describes the interpretation of p-values, which are provided in the PCC output.

Table 3. Common P-values and Interpretations

p-value	Interpretation
< 0.001	Extremely strong evidence against the null hypothesis.
< 0.01	Very strong evidence against the null hypothesis.
< 0.05	Moderate evidence against the null; commonly considered statistically significant.
0.05 – 0.10	Weak evidence against the null; sometimes called marginal or trend-level.
> 0.10	No statistically significant evidence against the null hypothesis.

Note that the null hypothesis is a default assumption in statistical testing that there is no effect, no difference, or no relationship between variables.

IV. Unadjusted and Partial Correlations

Now, suppose the unadjusted correlation output is statistically significant, but the partial correlation output is not. This suggests that the initial relationship between the variables of interest was distorted by extraneous variables controlled for via partial correlation. In this case, the PCC revealed a more accurate measure of association.

In the case that both the unadjusted and partial correlation output is similar in magnitude and statistical significance, the user may infer that the correlation between the variables of interest is not being distorted by the extraneous variables controlled for via partial correlation.

IV. Data Output

After analysis has been completed, users may **Download** their results (Figure 1, Item 10) for later reference.

The screenshot displays the 'PCC | Partial Correlation Calculator' interface. The top navigation bar includes 'PCC | Partial Correlation Calculator', 'Calculator', and 'Information'. The main panel is divided into two sections: a left sidebar for input and a right section for 'Analysis Output'.

Input Section (Left Sidebar):

- 1.** Select correlation type: Spearman (Default) [dropdown]
- 2.** Select significance level: 0.05 (Default) [dropdown]
- 3.** Upload dataset (.csv, .xls, .xlsx): [Browse... button, No file selected text]
- 4.** ☒ My data has headers
- 5.** Choose X and Y Variables: [text input]
- 6.** Choose Z Continuous Variable(s): [text input]
- 7.** Choose Z Categorical Variable(s): [text input]
- 8.** Analyze [button]
- 9.** Reset [button]
- 10.** Output available for download following analysis.

Analysis Output Section (Right):

- 11.** Unadjusted Correlation: 11.
- 12.** Partial Correlation: 12.
- 13.** Uploaded Data: 13.

Figure 1. **Calculator** panel of the PCC.

The PCC does not store analytic output between sessions, so users are encouraged to download their results. The **Reset** button (Figure 1, Item 9) is used to reset the calculator entirely. Results must be downloaded before resetting, or else they will be lost.